🦠 COVID-19 Mexico Dataset — Exploratory Data Analysis (EDA)

Internship Task 2 | EDA on Real-World Healthcare Data

📌 About the Dataset

This dataset contains COVID-19 patient records from Mexico, including demographic information, medical conditions, and outcomes. It contains ~1 million records.

Key Variables:

Column	Description
USMER	Treated in USMER units (1=Yes, 2=No)
MEDICAL_UNIT	Type of medical unit
SEX	1=Female, 2=Male
PATIENT_TYPE	1=Returned home, 2=Hospitalized
DATE_DIED	Date of death (9999-99-99 = survived)
INTUBED	Was patient intubated? (1=Yes, 2=No, 97=N/A)
PNEUMONIA	Has pneumonia?
AGE	Age of patient
PREGNANT	Is patient pregnant?
DIABETES	Has diabetes?
COPD	Has COPD?
ASTHMA	Has asthma?
INMSUPR	Immunosuppressed?
HIPERTENSION	Has hypertension?
OTHER_DISEASE	Other disease?
CARDIOVASCULAR	Cardiovascular disease?
OBESITY	Is obese?
RENAL_CHRONIC	Chronic renal disease?
TOBACCO	Tobacco user?
CLASIFFICATION_FINAL	COVID test result (1-3=Positive, 4-7=Negative/Other)

Column	Description
ICU	Admitted to ICU?

🤔 Step 1: Meaningful Questions Before Analysis

Before diving in, let's ask important questions:

1. **Mortality:** What is the overall death rate in this dataset?
2. **Demographics:** How does age and sex affect COVID-19 outcomes?
3. **Comorbidities:** Which pre-existing conditions (diabetes, hypertension, obesity) are most common in severe/fatal cases?
4. **Hospitalization:** What factors predict whether a patient is hospitalized vs. sent home?
5. **ICU & Intubation:** What percentage of patients needed ICU or intubation?
6. **Data Quality:** Are there missing values, invalid codes (97, 98, 99), or anomalies?
7. **Trends:** Is there a time trend in the number of cases or deaths?

📦 Step 2: Import Libraries

In [6]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import warnings

warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid', palette='muted')
plt.rcParams['figure.figsize'] = (12, 5)

print('✅ Libraries loaded successfully!')
```

```
✅ Libraries loaded successfully!
```

📂 Step 3: Load the Dataset

In [7]:
```
# Load compressed CSV
```

```
df = pd.read_csv("E:\Ineternship\Covid Data.csv")

print(f'✅ Dataset loaded!')
print(f'📊 Shape: {df.shape[0]:,} rows × {df.shape[1]} columns')
df.head()
```

```
✅ Dataset loaded!
📊 Shape: 1,048,575 rows × 21 columns
```

Out[7]:

	USMER	MEDICAL_UNIT	SEX	PATIENT_TYPE	DATE_DIED	INTUBED	PNEUMON
0	2	1	1	1	03/05/2020	97	
1	2	1	2	1	03/06/2020	97	
2	2	1	2	2	09/06/2020	1	
3	2	1	1	1	12/06/2020	97	
4	2	1	2	1	21/06/2020	97	

5 rows × 21 columns

🔍 Step 4: Data Structure Exploration

In [8]:

```
print('=== DATA TYPES ===')
print(df.dtypes)
print()
print('=== SHAPE ===')
print(f'Rows: {df.shape[0]:,} | Columns: {df.shape[1]}')
```

```
=== DATA TYPES ===
USMER                    int64
MEDICAL_UNIT             int64
SEX                      int64
PATIENT_TYPE             int64
DATE_DIED               object
INTUBED                  int64
PNEUMONIA                int64
AGE                      int64
PREGNANT                 int64
DIABETES                 int64
COPD                     int64
ASTHMA                   int64
INMSUPR                  int64
HIPERTENSION             int64
OTHER_DISEASE            int64
CARDIOVASCULAR           int64
OBESITY                  int64
RENAL_CHRONIC            int64
TOBACCO                  int64
CLASIFFICATION_FINAL     int64
ICU                      int64
dtype: object

=== SHAPE ===
Rows: 1,048,575 | Columns: 21
```

In [9]:

```python
print('=== BASIC STATISTICS ===')
df.describe()
```

```
=== BASIC STATISTICS ===
```

Out[9]:

	USMER	MEDICAL_UNIT	SEX	PATIENT_TYPE	INTUBED
count	1.048575e+06	1.048575e+06	1.048575e+06	1.048575e+06	1.048575e+06
mean	1.632194e+00	8.980565e+00	1.499259e+00	1.190765e+00	7.952288e+01
std	4.822084e-01	3.723278e+00	4.999997e-01	3.929041e-01	3.686889e+01
min	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
25%	1.000000e+00	4.000000e+00	1.000000e+00	1.000000e+00	9.700000e+01
50%	2.000000e+00	1.200000e+01	1.000000e+00	1.000000e+00	9.700000e+01
75%	2.000000e+00	1.200000e+01	2.000000e+00	1.000000e+00	9.700000e+01
max	2.000000e+00	1.300000e+01	2.000000e+00	2.000000e+00	9.900000e+01

In [10]:

```python
# Check missing values
print('=== MISSING VALUES ===')
missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)
missing_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct}
print(missing_df[missing_df['Missing Count'] > 0])
```

```
if missing.sum() == 0:
    print('✅ No null values found! (Note: Missing data may be encoded as 97,
```

```
=== MISSING VALUES ===
Empty DataFrame
Columns: [Missing Count, Missing %]
Index: []
✅ No null values found! (Note: Missing data may be encoded as 97, 98, 99)
```

In [11]:

```
# Check special codes: 97=N/A, 98=Ignored, 99=Missing
special_cols = ['INTUBED', 'PREGNANT', 'ICU']
print('=== SPECIAL CODES (97/98/99) ===')
for col in special_cols:
    if col in df.columns:
        count_97 = (df[col] == 97).sum()
        count_98 = (df[col] == 98).sum()
        count_99 = (df[col] == 99).sum()
        print(f'{col}: 97={count_97:,} | 98={count_98:,} | 99={count_99:,}')
```

```
=== SPECIAL CODES (97/98/99) ===
INTUBED: 97=848,544 | 98=0 | 99=7,325
PREGNANT: 97=523,511 | 98=3,754 | 99=0
ICU: 97=848,544 | 98=0 | 99=7,488
```

🧹 Step 5: Data Cleaning & Feature Engineering

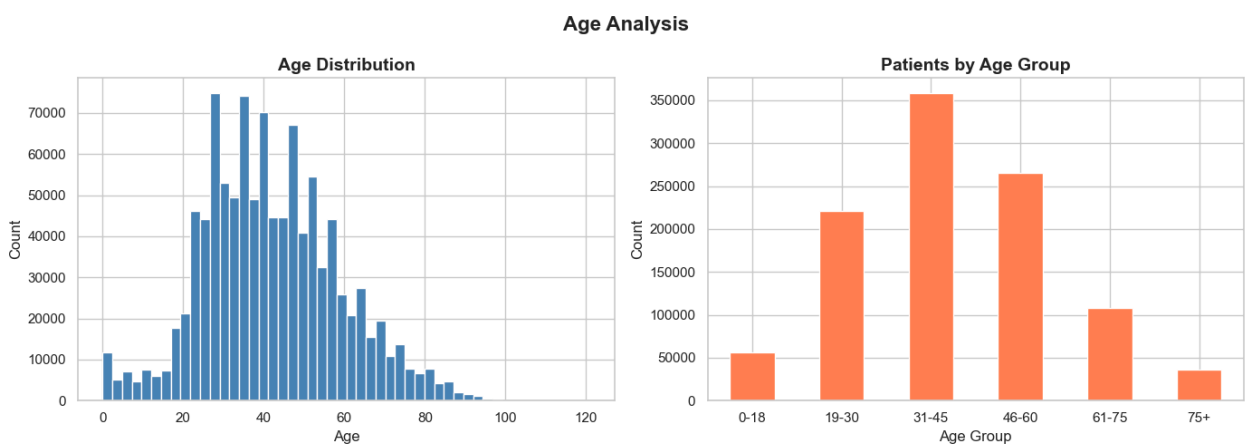
In [12]:

```
# Create DIED column: if DATE_DIED is a real date (not '9999-99-99'), patient
df['DIED'] = df['DATE_DIED'].apply(lambda x: 0 if str(x).strip() == '9999-99-9

# Convert DATE_DIED to datetime (keep only real dates)
df['DATE_DIED_PARSED'] = pd.to_datetime(df['DATE_DIED'], errors='coerce')

# COVID Positive: CLASIFFICATION_FINAL 1, 2, 3 = Positive
df['COVID_POSITIVE'] = df['CLASIFFICATION_FINAL'].apply(lambda x: 1 if x in [1

# Age groups
bins = [0, 18, 30, 45, 60, 75, 120]
labels = ['0-18', '19-30', '31-45', '46-60', '61-75', '75+']
df['AGE_GROUP'] = pd.cut(df['AGE'], bins=bins, labels=labels, right=True)

# Sex label
df['SEX_LABEL'] = df['SEX'].map({1: 'Female', 2: 'Male'})

print(f'✅ Feature engineering done!')
print(f'Deaths: {df["DIED"].sum():,} | Survived: {(df["DIED"]==0).sum():,}')
print(f'COVID Positive: {df["COVID_POSITIVE"].sum():,}')
```

```
✅ Feature engineering done!
Deaths: 76,942 | Survived: 971,633
COVID Positive: 391,979
```

📊 Step 6: Univariate Analysis

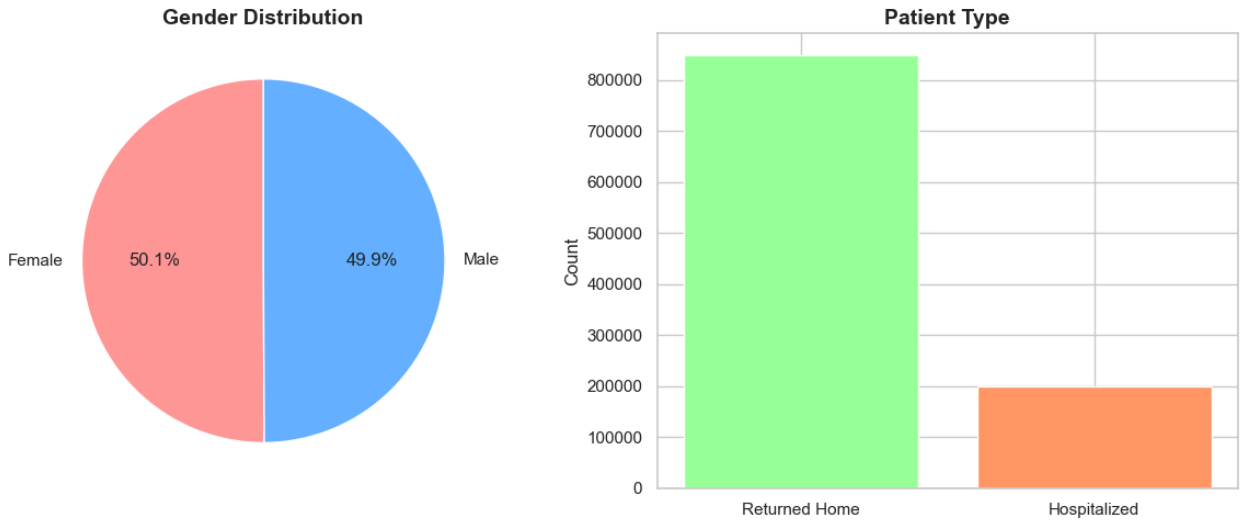
In [13]:
```
# AGE Distribution
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].hist(df['AGE'], bins=50, color='steelblue', edgecolor='white')
axes[0].set_title('Age Distribution', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Age')
axes[0].set_ylabel('Count')

df['AGE_GROUP'].value_counts().sort_index().plot(kind='bar', ax=axes[1], color
axes[1].set_title('Patients by Age Group', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Age Group')
axes[1].set_ylabel('Count')
axes[1].tick_params(axis='x', rotation=0)

plt.suptitle('Age Analysis', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()
print(f'Mean Age: {df["AGE"].mean():.1f} | Median: {df["AGE"].median():.0f} |
```



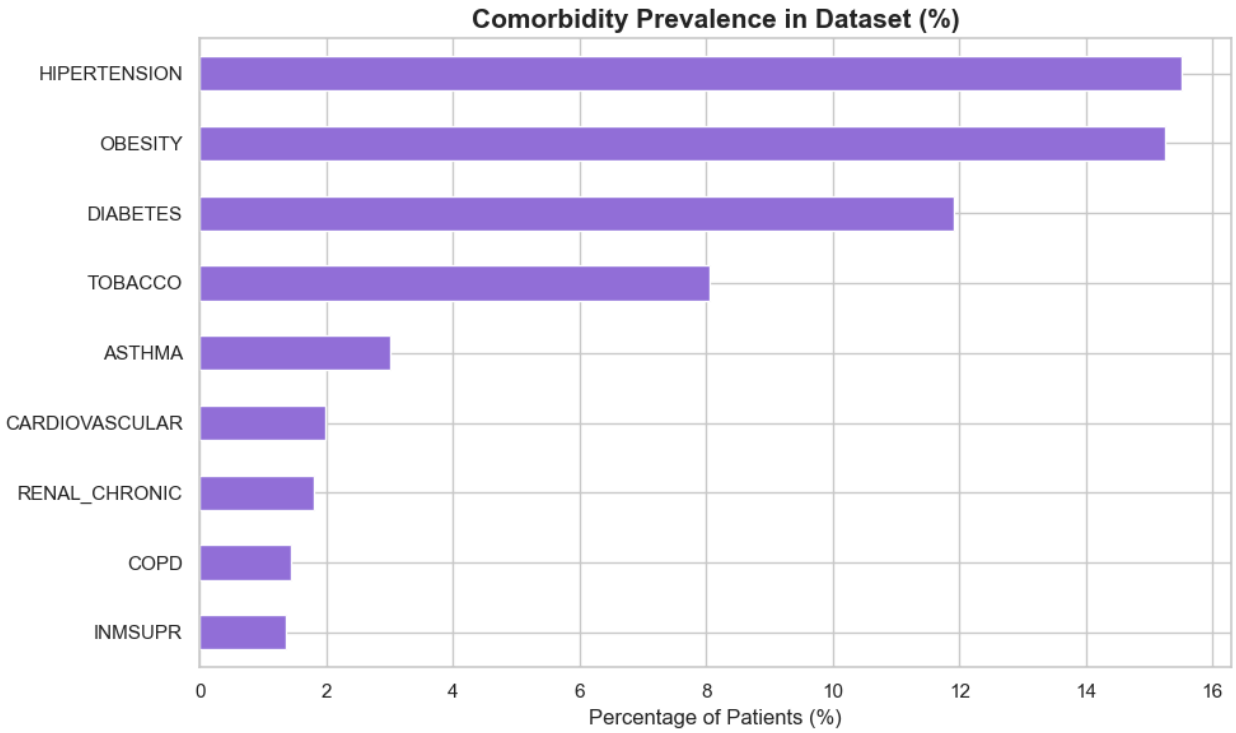

```
Mean Age: 41.8 | Median: 40 | Std: 16.9
```

In [14]:
```
# Sex & Patient Type Distribution
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sex_counts = df['SEX_LABEL'].value_counts()
axes[0].pie(sex_counts, labels=sex_counts.index, autopct='%1.1f%%',
            colors=['#ff9999', '#66b3ff'], startangle=90)
axes[0].set_title('Gender Distribution', fontsize=14, fontweight='bold')

pt_counts = df['PATIENT_TYPE'].value_counts()
pt_labels = {1: 'Returned Home', 2: 'Hospitalized'}
axes[1].bar([pt_labels.get(k, str(k)) for k in pt_counts.index],
            pt_counts.values, color=['#99ff99', '#ff9966'])
axes[1].set_title('Patient Type', fontsize=14, fontweight='bold')
axes[1].set_ylabel('Count')
```

```
plt.tight_layout()
plt.show()
```

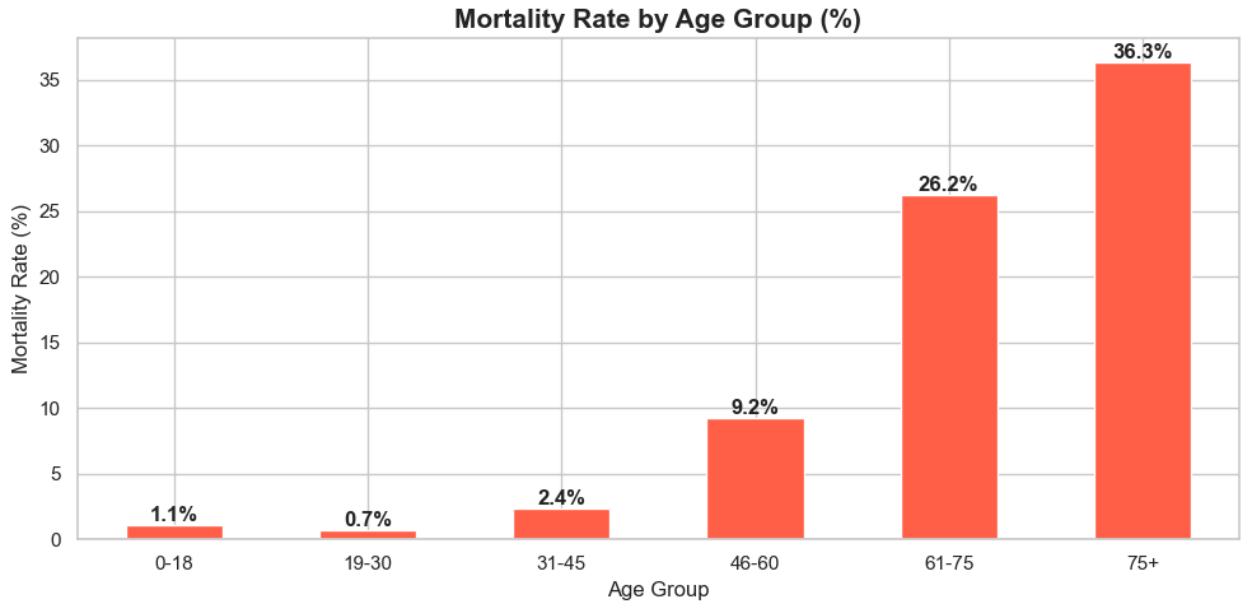


In [15]:
```
# Comorbidities Prevalence
comorbidities = ['DIABETES', 'HIPERTENSION', 'OBESITY', 'CARDIOVASCULAR',
                 'COPD', 'ASTHMA', 'RENAL_CHRONIC', 'TOBACCO', 'INMSUPR']

# 1 = Has condition, 2 = Does not have
comorbidity_rates = {}
for col in comorbidities:
    if col in df.columns:
        rate = (df[col] == 1).sum() / len(df) * 100
        comorbidity_rates[col] = rate

comorbidity_df = pd.Series(comorbidity_rates).sort_values(ascending=True)

plt.figure(figsize=(10, 6))
comorbidity_df.plot(kind='barh', color='mediumpurple', edgecolor='white')
plt.title('Comorbidity Prevalence in Dataset (%)', fontsize=15, fontweight='bo
plt.xlabel('Percentage of Patients (%)')
plt.tight_layout()
plt.show()
```



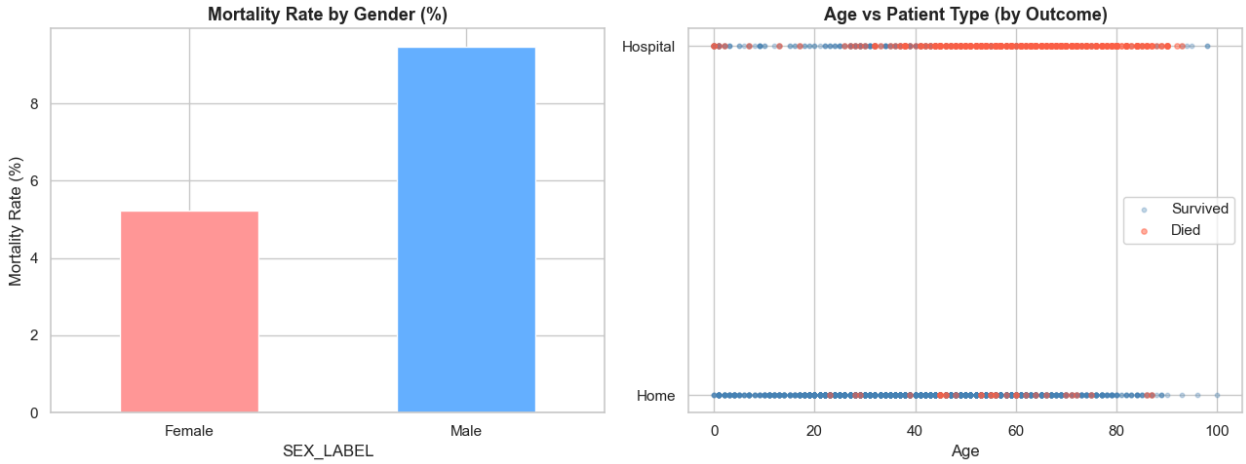
📈 Step 7: Bivariate Analysis — Mortality & Risk Factors

In [16]:
```
# Mortality Rate by Age Group
mortality_age = df.groupby('AGE_GROUP')['DIED'].mean() * 100

plt.figure(figsize=(10, 5))
mortality_age.plot(kind='bar', color='tomato', edgecolor='white')
plt.title('Mortality Rate by Age Group (%)', fontsize=15, fontweight='bold')
plt.xlabel('Age Group')
plt.ylabel('Mortality Rate (%)')
plt.xticks(rotation=0)
for i, v in enumerate(mortality_age):
    plt.text(i, v + 0.3, f'{v:.1f}%', ha='center', fontweight='bold')
plt.tight_layout()
plt.show()
```

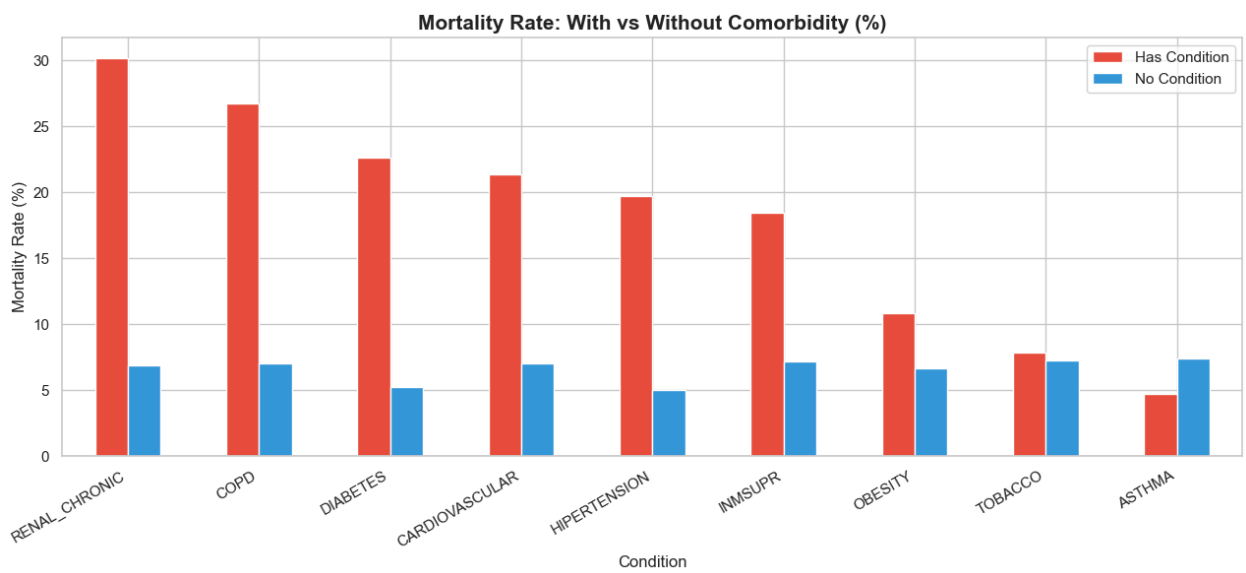


In [17]:
```
# Mortality Rate by Sex
mortality_sex = df.groupby('SEX_LABEL')['DIED'].mean() * 100
print('=== MORTALITY RATE BY SEX ===')
print(mortality_sex.to_string())

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

mortality_sex.plot(kind='bar', ax=axes[0], color=['#ff9999', '#66b3ff'], edgec
axes[0].set_title('Mortality Rate by Gender (%)', fontsize=13, fontweight='bol
axes[0].set_ylabel('Mortality Rate (%)')
axes[0].tick_params(axis='x', rotation=0)

# Age vs Mortality (scatter with hospitalization)
sample = df.sample(5000, random_state=42)
axes[1].scatter(sample[sample['DIED']==0]['AGE'], sample[sample['DIED']==0]['P
                alpha=0.3, color='steelblue', label='Survived', s=10)
axes[1].scatter(sample[sample['DIED']==1]['AGE'], sample[sample['DIED']==1]['P
                alpha=0.5, color='tomato', label='Died', s=15)
axes[1].set_title('Age vs Patient Type (by Outcome)', fontsize=13, fontweight=
axes[1].set_xlabel('Age')
axes[1].set_yticks([1, 2])
axes[1].set_yticklabels(['Home', 'Hospital'])
axes[1].legend()

plt.tight_layout()
plt.show()
```

```
=== MORTALITY RATE BY SEX ===
SEX_LABEL
Female    5.218792
Male      9.463029
```

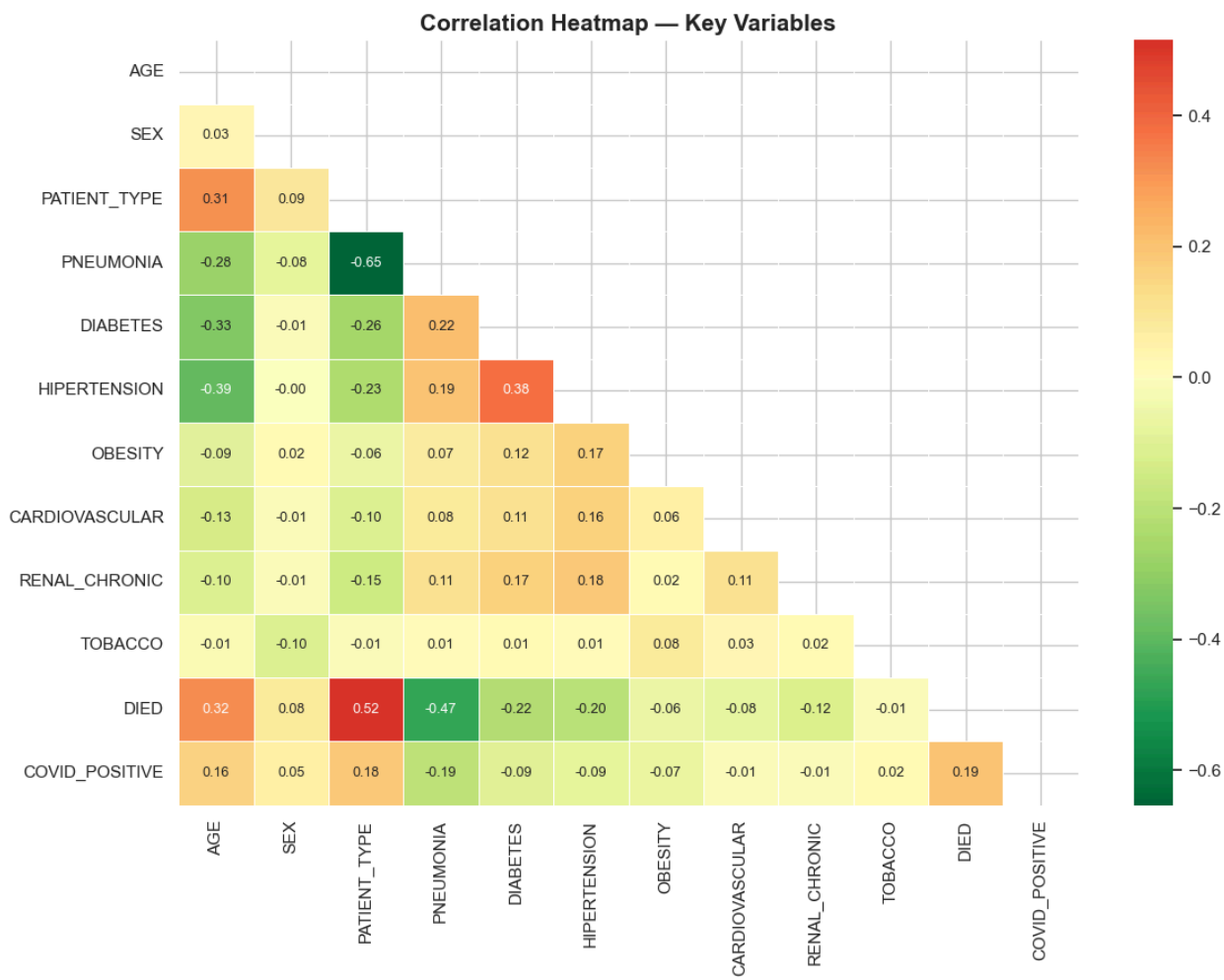


In [18]:
```
# Mortality by Comorbidity
comorbidity_mortality = {}
for col in comorbidities:
    if col in df.columns:
        has_condition = df[df[col] == 1]['DIED'].mean() * 100
        no_condition = df[df[col] == 2]['DIED'].mean() * 100
        comorbidity_mortality[col] = {'Has Condition': has_condition, 'No Conc

comorbidity_mort_df = pd.DataFrame(comorbidity_mortality).T.sort_values('Has C

comorbidity_mort_df.plot(kind='bar', figsize=(13, 6), color=['#e74c3c', '#3498
plt.title('Mortality Rate: With vs Without Comorbidity (%)', fontsize=15, font
plt.xlabel('Condition')
plt.ylabel('Mortality Rate (%)')
plt.xticks(rotation=30, ha='right')
plt.legend()
plt.tight_layout()
plt.show()
```



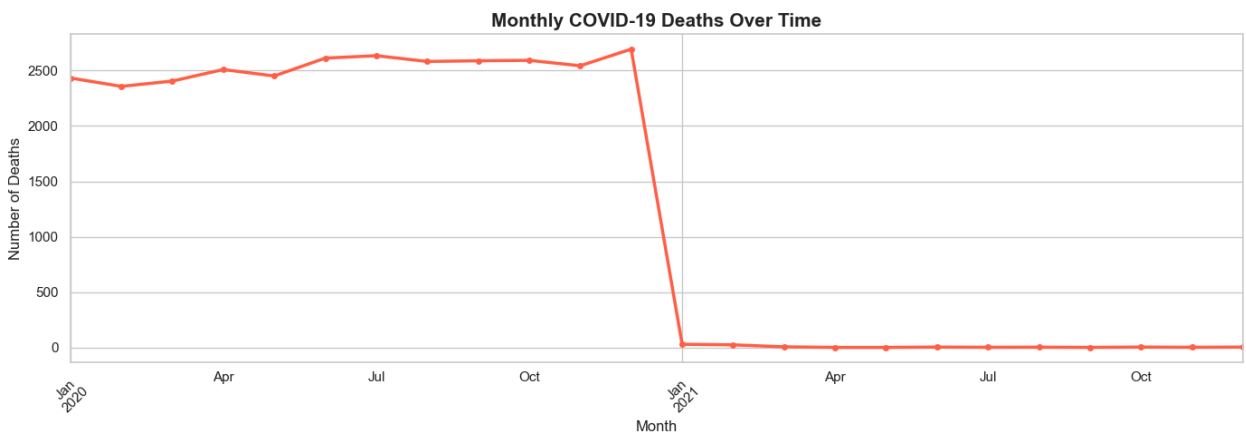
🔥 Step 8: Correlation & Heatmap Analysis

In [19]:
```
# Correlation heatmap for key features
key_features = ['AGE', 'SEX', 'PATIENT_TYPE', 'PNEUMONIA', 'DIABETES',
                'HIPERTENSION', 'OBESITY', 'CARDIOVASCULAR', 'RENAL_CHRONIC',
                'TOBACCO', 'DIED', 'COVID_POSITIVE']

corr_df = df[key_features].copy()
# Replace 97/98/99 with NaN for correlation
corr_df = corr_df.replace([97, 98, 99], np.nan)

plt.figure(figsize=(12, 9))
mask = np.triu(np.ones_like(corr_df.corr(), dtype=bool))
sns.heatmap(corr_df.corr(), mask=mask, annot=True, fmt='.2f',
            cmap='RdYlGn_r', center=0, linewidths=0.5,
            annot_kws={'size': 9})
plt.title('Correlation Heatmap — Key Variables', fontsize=15, fontweight='bolc
plt.tight_layout()
plt.show()
```



📅 Step 9: Time Trend Analysis

In [20]:
```
# Deaths over time
deaths_over_time = df[df['DIED'] == 1]['DATE_DIED_PARSED'].dropna()
deaths_monthly = deaths_over_time.dt.to_period('M').value_counts().sort_index(

plt.figure(figsize=(14, 5))
deaths_monthly.plot(kind='line', color='tomato', linewidth=2.5, marker='o', ma
plt.title('Monthly COVID-19 Deaths Over Time', fontsize=15, fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Number of Deaths')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



🧪 Step 10: Hypothesis Testing

In [21]:
```
from scipy import stats

# Hypothesis 1: Do patients who died have significantly higher age?
died_ages = df[df['DIED'] == 1]['AGE'].dropna()
survived_ages = df[df['DIED'] == 0]['AGE'].dropna()

t_stat, p_value = stats.ttest_ind(died_ages, survived_ages)
print('=== HYPOTHESIS TEST 1: Age vs Mortality ===')
print(f'Mean Age (Died):     {died_ages.mean():.1f}')
print(f'Mean Age (Survived): {survived_ages.mean():.1f}')
print(f'T-statistic: {t_stat:.3f} | P-value: {p_value:.5e}')
if p_value < 0.05:
    print('✅ RESULT: Statistically significant difference — older patients ha
else:
    print('❌ RESULT: No significant difference found.')
```

```
=== HYPOTHESIS TEST 1: Age vs Mortality ===
Mean Age (Died):     61.1
Mean Age (Survived): 40.3
T-statistic: 346.831 | P-value: 0.00000e+00
✅ RESULT: Statistically significant difference — older patients have higher mo
rtality.
```

In [22]:

```python
# Hypothesis 2: Chi-Square test — Diabetes vs Mortality
from scipy.stats import chi2_contingency

# Filter for valid values only (1=Yes, 2=No)
valid = df[df['DIABETES'].isin([1, 2])]
contingency_table = pd.crosstab(valid['DIABETES'], valid['DIED'])
contingency_table.index = ['Has Diabetes', 'No Diabetes']
contingency_table.columns = ['Survived', 'Died']

chi2, p, dof, expected = chi2_contingency(contingency_table)

print('=== HYPOTHESIS TEST 2: Diabetes vs Mortality (Chi-Square) ===')
print(contingency_table)
print(f'\nChi2: {chi2:.2f} | P-value: {p:.5e} | DOF: {dof}')
if p < 0.05:
    print('✅ RESULT: Significant association between diabetes and mortality.'
else:
    print('❌ No significant association.')
```

```
=== HYPOTHESIS TEST 2: Diabetes vs Mortality (Chi-Square) ===
              Survived   Died
Has Diabetes     96724  28265
No Diabetes     872302  47946

Chi2: 49306.88 | P-value: 0.00000e+00 | DOF: 1
✅ RESULT: Significant association between diabetes and mortality.
```

In [23]:

```python
# Hypothesis 3: Is hospitalization rate different for COVID-positive vs negati
pos = df[df['COVID_POSITIVE'] == 1]['PATIENT_TYPE']
neg = df[df['COVID_POSITIVE'] == 0]['PATIENT_TYPE']

hosp_pos = (pos == 2).mean() * 100
hosp_neg = (neg == 2).mean() * 100

print('=== HYPOTHESIS TEST 3: COVID+ vs Hospitalization ===')
print(f'Hospitalization Rate (COVID+): {hosp_pos:.1f}%')
print(f'Hospitalization Rate (COVID-): {hosp_neg:.1f}%')

cont = pd.crosstab(df['COVID_POSITIVE'], df['PATIENT_TYPE'])
chi2_h, p_h, _, _ = chi2_contingency(cont)
print(f'Chi2: {chi2_h:.2f} | P-value: {p_h:.5e}')
if p_h < 0.05:
    print('✅ RESULT: COVID+ patients have significantly different hospitaliza
```

```
=== HYPOTHESIS TEST 3: COVID+ vs Hospitalization ===
Hospitalization Rate (COVID+): 28.4%
Hospitalization Rate (COVID-): 13.5%
Chi2: 35190.62 | P-value: 0.00000e+00
✅ RESULT: COVID+ patients have significantly different hospitalization rates.
```

⚠ Step 11: Anomaly & Data Quality Detection

In [24]:

```
print('=== DATA QUALITY REPORT ===')

# Age outliers
age_outliers = df[(df['AGE'] > 110) | (df['AGE'] < 0)]
print(f'\n1. Age Outliers (< 0 or > 110): {len(age_outliers)}')

# Age distribution check
print(f'   Min Age: {df["AGE"].min()} | Max Age: {df["AGE"].max()}')

# Check pregnant males
pregnant_males = df[(df['SEX'] == 2) & (df['PREGNANT'] == 1)]
print(f'\n2. Pregnant Males (anomaly): {len(pregnant_males)}')

# Duplicate rows
duplicates = df.duplicated().sum()
print(f'\n3. Duplicate Rows: {duplicates}')

# Invalid classification codes
valid_classification = df['CLASIFFICATION_FINAL'].isin(range(1, 8))
invalid_class = (~valid_classification).sum()
print(f'\n4. Invalid CLASIFFICATION_FINAL codes: {invalid_class}')

# Summary of special codes across binary features
binary_cols = ['PNEUMONIA', 'DIABETES', 'COPD', 'ASTHMA', 'INMSUPR',
               'HIPERTENSION', 'OTHER_DISEASE', 'CARDIOVASCULAR',
               'OBESITY', 'RENAL_CHRONIC', 'TOBACCO']
print('\n5. Special Code Presence (97/98/99) in Binary Columns:')
for col in binary_cols:
    if col in df.columns:
        special = df[col].isin([97, 98, 99]).sum()
        if special > 0:
            print(f'   {col}: {special:,} special codes ({special/len(df)*100:

print('\n✅ Data Quality Check Complete!')
```

```
=== DATA QUALITY REPORT ===

1. Age Outliers (< 0 or > 110): 22
   Min Age: 0 | Max Age: 121

2. Pregnant Males (anomaly): 0

3. Duplicate Rows: 812049

4. Invalid CLASIFFICATION_FINAL codes: 0

5. Special Code Presence (97/98/99) in Binary Columns:
   PNEUMONIA: 16,003 special codes (1.53%)
   DIABETES: 3,338 special codes (0.32%)
   COPD: 3,003 special codes (0.29%)
   ASTHMA: 2,979 special codes (0.28%)
   INMSUPR: 3,404 special codes (0.32%)
   HIPERTENSION: 3,104 special codes (0.30%)
   OTHER_DISEASE: 5,045 special codes (0.48%)
   CARDIOVASCULAR: 3,076 special codes (0.29%)
   OBESITY: 3,032 special codes (0.29%)
   RENAL_CHRONIC: 3,006 special codes (0.29%)
   TOBACCO: 3,220 special codes (0.31%)

✅ Data Quality Check Complete!
```

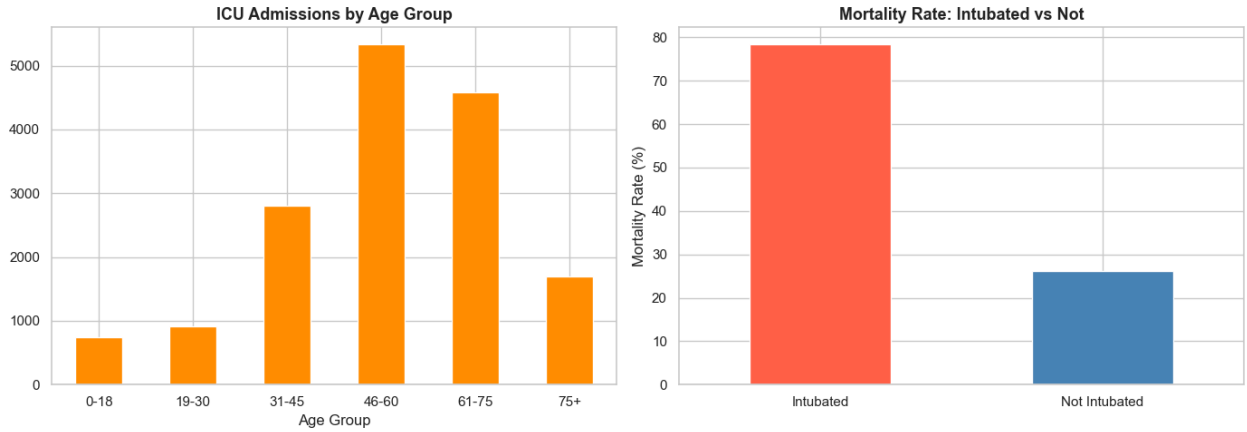
🏥 Step 12: ICU & Intubation Analysis

In [25]:

```
# ICU & Intubation rates (among hospitalized patients only)
hospitalized = df[df['PATIENT_TYPE'] == 2]

icu_rate = (hospitalized['ICU'] == 1).sum() / len(hospitalized) * 100
intubated_rate = (hospitalized['INTUBED'] == 1).sum() / len(hospitalized) * 10

print(f'=== ICU & INTUBATION (Hospitalized Patients Only) ===')
print(f'Total Hospitalized: {len(hospitalized):,}')
print(f'ICU Admission Rate: {icu_rate:.2f}%')
print(f'Intubation Rate:    {intubated_rate:.2f}%')

# ICU by age group
icu_age = hospitalized[hospitalized['ICU'] == 1].groupby('AGE_GROUP').size()

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
icu_age.plot(kind='bar', ax=axes[0], color='darkorange', edgecolor='white')
axes[0].set_title('ICU Admissions by Age Group', fontsize=13, fontweight='bolc
axes[0].set_xlabel('Age Group')
axes[0].tick_params(axis='x', rotation=0)

# Intubation vs Outcome
intub_outcome = df[df['INTUBED'].isin([1, 2])].groupby('INTUBED')['DIED'].mear
```

```
intub_outcome.index = ['Intubated', 'Not Intubated']
intub_outcome.plot(kind='bar', ax=axes[1], color=['tomato', 'steelblue'], edge
axes[1].set_title('Mortality Rate: Intubated vs Not', fontsize=13, fontweight=
axes[1].set_ylabel('Mortality Rate (%)')
axes[1].tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.show()
```

```
=== ICU & INTUBATION (Hospitalized Patients Only) ===
Total Hospitalized: 200,031
ICU Admission Rate: 8.43%
Intubation Rate:    16.83%
```



📝 Step 13: EDA Summary & Key Findings

In [26]:
```
overall_mortality = df['DIED'].mean() * 100
hosp_rate = (df['PATIENT_TYPE'] == 2).mean() * 100
covid_pos_rate = df['COVID_POSITIVE'].mean() * 100

print('=' * 60)
print('          EDA SUMMARY — COVID-19 MEXICO DATASET')
print('=' * 60)
print(f'  Total Records:          {len(df):,}')
print(f'  Overall Mortality Rate: {overall_mortality:.2f}%')
print(f'  Hospitalization Rate:   {hosp_rate:.2f}%')
print(f'  COVID Positive Rate:    {covid_pos_rate:.2f}%')
print()
print('KEY FINDINGS:')
print('  1. Older patients (60+) have significantly higher mortality')
print('  2. Males have higher mortality than females')
print('  3. Pneumonia, Diabetes & Hypertension are most common comorbidities')
print('  4. Comorbidities (especially renal, cardiovascular) raise death risk'
print('  5. Intubated patients have much higher mortality (critical condition)
print('  6. Some anomalies found: pregnant males, special codes (97/98/99)')
print('  7. COVID+ patients are significantly more likely to be hospitalized')
print()
print('RECOMMENDATIONS FOR FURTHER ANALYSIS:')
```

```
print('  - Build a mortality prediction model (Logistic Regression / Random Fo
print('  - Handle special codes (97/98/99) properly before modeling')
print('  - Investigate seasonal/wave patterns in the death trend')
print('  - Analyze interaction effects between multiple comorbidities')
print('=' * 60)
```

```
============================================================
         EDA SUMMARY — COVID-19 MEXICO DATASET
============================================================
  Total Records:          1,048,575
  Overall Mortality Rate: 7.34%
  Hospitalization Rate:   19.08%
  COVID Positive Rate:    37.38%

KEY FINDINGS:
  1. Older patients (60+) have significantly higher mortality
  2. Males have higher mortality than females
  3. Pneumonia, Diabetes & Hypertension are most common comorbidities
  4. Comorbidities (especially renal, cardiovascular) raise death risk
  5. Intubated patients have much higher mortality (critical condition)
  6. Some anomalies found: pregnant males, special codes (97/98/99)
  7. COVID+ patients are significantly more likely to be hospitalized

RECOMMENDATIONS FOR FURTHER ANALYSIS:
  - Build a mortality prediction model (Logistic Regression / Random Forest)
  - Handle special codes (97/98/99) properly before modeling
  - Investigate seasonal/wave patterns in the death trend
  - Analyze interaction effects between multiple comorbidities
============================================================
```

EDA Notebook — COVID-19 Mexico Dataset | Internship Task 2